

On the Theory and Algebraic Properties of Determinants

An Axiomatic and Inductive Development of Row Operations and n-Linearity

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Abstract

This paper presents a self-contained, rigorous exposition of the determinant function defined on the space of square matrices $M_{n \times n}(F)$ over a field F . Beginning with the recursive definition via cofactor expansion along the first row, we prove the fundamental multilinear and alternating properties of determinants under elementary row operations. We analyze the behavior of determinants under matrix multiplication, and demonstrate that $\det(A^T) = \det(A)$, thereby extending all row-operation properties to column operations.

1 Introduction

The determinant, which has played a prominent role in the theory of linear algebra, is a special scalar-valued function defined on the set of square matrices. Although the determinant is not a linear transformation on $M_{n \times n}(F)$ for $n > 1$, it does possess a kind of linearity (called n-linearity) as well as other properties that are examined in this paper.

2 Determinants of order n

Given $A \in M_{n \times n}(F)$, for $n \geq 2$, denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j by $\tilde{A}_{ij}$.

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad \tilde{A}_{11} = \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}, \quad \tilde{A}_{23} = \begin{bmatrix} 1 & 2 \\ 7 & 8 \end{bmatrix}$$

Definition 2.1. Let $A \in M_{n \times n}(F)$. If $n = 1$ then, $A = [A_{11}]$ and $\det(A) = A_{11}$.

For $n \geq 2$, we define $\det(A)$ recursively as

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} A_{1j} \cdot \det(\tilde{A}_{1j}).$$

The scalar $\det(A)$ is called the **determinant** of A and is also denoted by $|A|$.

Definition 2.2. The scalar

$$(-1)^{i+j} \det(\tilde{A}_{ij})$$

is called the **cofactor** of the entry of A in row i and column j .

Let us denote the cofactor of row i , column j entry of A then

$$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$$

and

$$\det(A) = A_{11}c_{11} + A_{12}c_{12} + \cdots + A_{1n}c_{1n}$$

Theorem 2.1. *The determinant of an $n \times n$ matrix is a linear function of each row when the remaining rows are held fixed. That is, for $1 \leq r \leq n$, we have*

$$\det \begin{pmatrix} a_1 \\ \vdots \\ a_{r-1} \\ u + kv \\ a_{r+1} \\ \vdots \\ a_n \end{pmatrix} = \det \begin{pmatrix} a_1 \\ \vdots \\ a_{r-1} \\ u \\ a_{r+1} \\ \vdots \\ a_n \end{pmatrix} + k \cdot \det \begin{pmatrix} a_1 \\ \vdots \\ a_{r-1} \\ v \\ a_{r+1} \\ \vdots \\ a_n \end{pmatrix}$$

whenever k is a scalar and u, v and each a_i are row vectors in F^n .

Proof. The proof is by mathematical induction on n . The result is immediate for $n = 1$. Now let us make an induction hypothesis that for some integer $n \geq 2$, the theorem holds true for $(n - 1) \times (n - 1)$ matrix. Let A be an $n \times n$ matrix with $a_1, a_2, \dots, a_n$ rows and suppose for some r ($1 \leq r \leq n$) we have $a_r = u + kv$ for some $u, v \in F^n$ and some scalar k . Let $u = (b_1, b_2, \dots, b_n)$ and $v = (c_1, c_2, \dots, c_n)$ and let B and C be the matrices obtained from A by replacing row r of A by u and v respectively. Now we need to prove

$$\det(A) = \det(B) + k \det(C).$$

For $r = 1$, since we change only the first row of A, B and C , then

$$\tilde{A}_{1j} = \tilde{B}_{1j} = \tilde{C}_{1j}$$

Now,

$$\begin{aligned}
\det(A) &= \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}) \\
&= \sum_{j=1}^n (-1)^{1+j} (b_j + kc_j) \det(\tilde{A}_{1j}) \\
&= \sum_{j=1}^n \left[(-1)^{1+j} b_j \det(\tilde{A}_{1j}) + k \cdot (-1)^{1+j} c_j \det(\tilde{A}_{1j}) \right] \\
&= \sum_{j=1}^n \left[(-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) + k(-1)^{1+j} C_{1j} \det(\tilde{C}_{1j}) \right] \\
&= \det(B) + k \det(C).
\end{aligned}$$

Now for $r > 1$ and $1 \leq j \leq n$, the rows of $\tilde{A}_{1j}$, $\tilde{B}_{1j}$ and $\tilde{C}_{1j}$ are same except for row $r - 1$. Also row $r - 1$ of $\tilde{A}_{1j}$ is

$$(b_1 + kc_1, \dots, b_{j-1} + kc_{j-1}, b_{j+1} + kc_{j+1}, \dots, b_n + kc_n),$$

which is the sum of row $r - 1$ of $\tilde{B}_{1j}$ and k times row $r - 1$ of $\tilde{C}_{1j}$.

Since $\tilde{B}_{1j}$ and $\tilde{C}_{1j}$ are $(n - 1) \times (n - 1)$ matrices, then from our induction hypothesis, we have

$$\det(\tilde{A}_{1j}) = \det(\tilde{B}_{1j}) + k \det(\tilde{C}_{1j}).$$

Also since $A_{1j} = B_{1j} = C_{1j}$, then

$$\begin{aligned}
\det(A) &= \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}) \\
&= \sum_{j=1}^n (-1)^{1+j} A_{1j} \left[\det(\tilde{B}_{1j}) + k \det(\tilde{C}_{1j}) \right] \\
&= \sum_{j=1}^n (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) + k \sum_{j=1}^n (-1)^{1+j} C_{1j} \det(\tilde{C}_{1j}) \\
&= \det(B) + k \det(C).
\end{aligned}$$

Thus the theorem is established for $n \times n$ matrices. □

Corollary 2.1.1. *If $A \in M_{n \times n}(F)$ has a row consisting entirely of zeroes, then $\det(A) = 0$.*

Proof. Let $A \in M_{n \times n}(F)$ be a matrix where the r -th row consists entirely of zeroes, such that $a_r = 0$. Thus we can write

$$a_r = 0 \cdot 0.$$

Thus,

$$\det(A) = \det \begin{pmatrix} a_1 \\ \vdots \\ a_{r-1} \\ 0 \cdot 0 \\ a_{r+1} \\ \vdots \\ a_n \end{pmatrix}$$

By Theorem 2.1,

$$\det(A) = 0 \cdot \det(A) = 0.$$

□

Lemma 2.1. *Let $B \in M_{n \times n}(F)$ where $n \geq 2$. If row i of B equals e_k for some k ($1 \leq k \leq n$), then $\det(B) = (-1)^{i+k} \det(\tilde{B}_{ik})$.*

Proof. The proof is by mathematical induction on n . Now let

$$B = \begin{bmatrix} 1 & 0 \\ a_3 & a_4 \end{bmatrix}$$

Then, $\det(B) = a_4 = (-1)^{1+1} \det(\tilde{B}_{11})$. Again let

$$B = \begin{bmatrix} a_1 & a_2 \\ 0 & 1 \end{bmatrix}$$

Then, $\det(B) = a_1 = (-1)^{2+2} \det(\tilde{B}_{22})$. Similarly we can verify the result for

$$B = \begin{bmatrix} 0 & 1 \\ a_3 & a_4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} a_1 & a_2 \\ 1 & 0 \end{bmatrix}.$$

Thus the theorem is true for $n = 2$.

Now let us take an induction hypothesis that for some integer $n \geq 3$ the theorem is true for $(n-1) \times (n-1)$ matrices and let B be an $n \times n$ matrix in which row i of B equals e_k for some k ($1 \leq k \leq n$).

For $i = 1$,

$$\begin{aligned} \det(B) &= \sum_{j=1}^n (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) \\ &= \sum_{j \neq k} (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) + \sum_{j=k} (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) \end{aligned}$$

Here we have taken that the first row of the matrix is e_k ($1 \leq k \leq n$) and also we are expanding the determinant for the first row. Thus if $j \neq k$ then $B_{1j} = 0$ and hence

$$\det(B) = 0 + (-1)^{1+k} \det(\tilde{B}_{1k}) = (-1)^{1+k} \det(\tilde{B}_{1k})$$

Thus the theorem holds for 1st row of B .

Suppose that $1 < i \leq n$. Then if we remove 1st row and j -th column of B , then the i -th row in B becomes $(i-1)$ -th row in $\tilde{B}_{1j}$. Now let C_{ij} denote the $(n-2) \times (n-2)$ matrix by removing $(i-1)$ -th row and k -th column of $\tilde{B}_{1j}$.

Now if we remove j -th column from B , then if $j < k$ then the k -th column becomes $(k-1)$ -th column in C_{ij} ; and if $j = k$ then the only column containing 1 in the i -th row in B is removed; and if $j > k$ then the k -th column remains intact. Thus by induction hypothesis and by Corollary 2.1.1,

$$\det(\tilde{B}_{1j}) = \begin{cases} (-1)^{(i-1)+(k-1)} \det(C_{ij}) & \text{if } j < k \\ 0 & \text{if } j = k \\ (-1)^{(i-1)+k} \det(C_{ij}) & \text{if } j > k \end{cases}$$

Thus,

$$\begin{aligned} \det(B) &= \sum_{j=1}^n (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) \\ &= \sum_{j < k} (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) + \sum_{j > k} (-1)^{1+j} B_{1j} \det(\tilde{B}_{1j}) \\ &= \sum_{j < k} (-1)^{1+j} B_{1j} [(-1)^{(i-1)+(k-1)} \det(C_{ij})] \\ &\quad + \sum_{j > k} (-1)^{1+j} B_{1j} [(-1)^{(i-1)+k} \det(C_{ij})] \\ &= \sum_{j < k} (-1)^{1+j} B_{1j} \cdot (-1)^{i+k} (-1)^{-2} \det(C_{ij}) \\ &\quad + \sum_{j > k} (-1)^{1+j} \cdot (-1)^{i+k} (-1)^{-1} B_{1j} \det(C_{ij}) \\ &= (-1)^{i+k} \left[\sum_{j < k} (-1)^{1+j} B_{1j} \det(C_{ij}) + \sum_{j > k} (-1)^{1+(j-1)} B_{1j} \det(C_{ij}) \right] \end{aligned}$$

Because the expression inside the above bracket is the cofactor expansion of $\tilde{B}_{ik}$ along first row. Thus,

$$\det(B) = (-1)^{i+k} \det(\tilde{B}_{ik}).$$

Thus the lemma holds true for all $n \times n$ matrices. □

Theorem 2.2. *The determinant of a square matrix can be evaluated by cofactor expansion along any row. That is, if $A \in M_{n \times n}(F)$, then for any integer i ($1 \leq i \leq n$),*

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$$

Proof. Cofactor expansion along first row of A gives the determinant of A by definition. So the result is true for $i = 1$. Fix $i > 1$. Row i of A can be written as $\sum_{j=1}^n A_{ij} e_j$. For $1 \leq j \leq n$, let B_j denote the

matrix obtained from A by replacing row i of A by e_j . Thus by Lemma 2.1,

$$\det(B_j) = (-1)^{i+j} \det(\tilde{B}_{ij}).$$

Since B_j is obtained from A by replacing row i of A with e_j , thus

$$\tilde{A}_{ij} = \tilde{B}_{ij}.$$

Therefore,

$$\det(B_j) = (-1)^{i+j} \det(\tilde{A}_{ij}).$$

Now,

$$\det(A) = \det \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_{i-1} \\ \sum_{j=1}^n A_{ij} e_j \\ R_{i+1} \\ \vdots \\ R_n \end{pmatrix} = \det \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_{i-1} \\ A_{i1}e_1 + A_{i2}e_2 + \cdots + A_{in}e_n \\ R_{i+1} \\ \vdots \\ R_n \end{pmatrix}$$

By Theorem 2.1,

$$\begin{aligned} \det(A) &= A_{i1} \det \begin{pmatrix} R_1 \\ \vdots \\ e_1 \\ \vdots \\ R_n \end{pmatrix} + A_{i2} \det \begin{pmatrix} R_1 \\ \vdots \\ e_2 \\ \vdots \\ R_n \end{pmatrix} + \cdots + A_{in} \det \begin{pmatrix} R_1 \\ \vdots \\ e_n \\ \vdots \\ R_n \end{pmatrix} \\ &= \sum_{j=1}^n A_{ij} \det(B_j) \\ &= \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}). \end{aligned}$$

□

Remark. The proof of this theorem tells us that not just the first row but we can find the value of the determinant by expanding any row in of the matrix.

Corollary 2.2.1. *If $A \in M_{n \times n}(F)$ has two identical rows, then $\det(A) = 0$.*

Proof. The proof is by mathematical induction on n . Let $A \in M_{2 \times 2}(F)$ such that the rows of A are identical. Let us say,

$$A = \begin{bmatrix} a_1 & a_2 \\ a_1 & a_2 \end{bmatrix}$$

Then

$$\det(A) = a_1 a_2 - a_1 a_2 = 0.$$

Thus the statement holds true for $n = 2$. Assume for some integer $n \geq 3$, the statement is true for $(n - 1) \times (n - 1)$ matrices, and let row r and s of $A \in M_{n \times n}(F)$ be identical for $r \neq s$. Because $n \geq 3$, we can choose some integer i ($1 \leq i \leq n$) other than r and s . Now by Theorem 2.2,

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Since $\tilde{A}_{ij}$ is an $(n - 1) \times (n - 1)$ matrix with two identical rows, by our induction hypothesis we can say that each $\det(\tilde{A}_{ij}) = 0$ and hence $\det(A) = 0$. Thus the statement holds true for every $n \times n$ matrix or every square matrix. $\square$

Theorem 2.3. *If $A \in M_{n \times n}(F)$ and B is a matrix obtained from A by interchanging any two rows of A , then $\det(B) = -\det(A)$.*

Proof. Let the rows of $A \in M_{n \times n}(F)$ be $a_1, a_2, \dots, a_n$, and let B be the matrix obtained from A by interchanging rows r and s , where $r < s$. Thus

$$A = \begin{pmatrix} a_1 \\ \vdots \\ a_r \\ \vdots \\ a_s \\ \vdots \\ a_n \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} a_1 \\ \vdots \\ a_s \\ \vdots \\ a_r \\ \vdots \\ a_n \end{pmatrix}$$

Consider the matrix obtained from A by replacing rows r and s by row $a_r + a_s$. By Corollary 2.2.1 and Theorem 2.1, we get,

$$\begin{aligned} & \det \begin{pmatrix} a_1 \\ \vdots \\ a_r + a_s \\ \vdots \\ a_r + a_s \\ \vdots \\ a_n \end{pmatrix} = 0. \\ \Rightarrow & \det \begin{pmatrix} a_1 \\ \vdots \\ a_r \\ \vdots \\ a_r \\ \vdots \\ a_n \end{pmatrix} + \det \begin{pmatrix} a_1 \\ \vdots \\ a_r \\ \vdots \\ a_s \\ \vdots \\ a_n \end{pmatrix} + \det \begin{pmatrix} a_1 \\ \vdots \\ a_s \\ \vdots \\ a_r \\ \vdots \\ a_n \end{pmatrix} + \det \begin{pmatrix} a_1 \\ \vdots \\ a_s \\ \vdots \\ a_s \\ \vdots \\ a_n \end{pmatrix} = 0 \\ \Rightarrow & 0 + \det(A) + \det(B) + 0 = 0 \end{aligned}$$

$$\Rightarrow \det(A) = -\det(B).$$

□

Theorem 2.4. Let $A \in M_{n \times n}(F)$, and let B be a matrix obtained by adding a multiple of one row of A to another row of A . Then $\det(B) = \det(A)$.

Proof. Suppose B is the $n \times n$ matrix obtained from A by adding k times row r to row s , where $r \neq s$. Let the rows of A be $a_1, a_2, \dots, a_n$ and rows of B be $b_1, b_2, \dots, b_n$. Then $b_i = a_i$ for $i \neq s$ and $b_s = a_s + ka_r$. Let C be the matrix obtained from A by replacing row s with a_r . Applying Theorem 2.1 to row s of B , we get

$$\det(B) = \det(A) + k \det(C) = \det(A)$$

because $\det(C) = 0$ by Corollary 2.2.1. □

Corollary 2.4.1. If $A \in M_{n \times n}(F)$ has rank less than n , then $\det(A) = 0$.

Proof. If the rank of A is less than n , then the rows $a_1, a_2, \dots, a_n$ of A are linearly dependent. Thus we can say, some row of A , say row r , is a linear combination of the other rows. So there exists scalars c_i such that

$$a_r = c_1 a_1 + c_2 a_2 + \dots + c_{r-1} a_{r-1} + c_{r+1} a_{r+1} + \dots + c_n a_n.$$

Let B be the matrix obtained from A by adding $-c_i$ times row i to row r for every $i \neq r$. Then, row r of B consists entirely of zeroes, and so $\det(B) = 0$. But by Theorem 2.4, $\det(A) = \det(B)$. Hence $\det(A) = 0$. □

Theorem 2.5. For any $A \in M_{n \times n}(F)$, $\det(kA) = k^n \det(A)$.

Proof. The proof is by mathematical induction on n . The result is immediate for $n = 1$. Now let us take our mathematical hypothesis that the theorem is true for any $(n-1) \times (n-1)$ matrix. Now let A be an $n \times n$ matrix then,

$$\det(kA) = \sum_{j=1}^n (-1)^{1+j} k(A)_{1j} \det(\widetilde{kA}_{1j}).$$

Now since $\widetilde{A}_{1j}$ is an $(n-1) \times (n-1)$ matrix, then by our hypothesis we can say that $\det(\widetilde{kA}_{1j}) = k^{n-1} \det(\widetilde{A}_{1j})$. Hence,

$$\begin{aligned} \det(kA) &= \sum_{j=1}^n (-1)^{1+j} kA_{1j} \left[k^{n-1} \det(\widetilde{A}_{1j}) \right] \\ &= k^n \sum_{j=1}^n (-1)^{1+j} A_{1j} \det(\widetilde{A}_{1j}) \\ &= k^n \det(A). \end{aligned}$$

Thus the statement holds true for $n \times n$ matrix. □

Theorem 2.6. If E is an elementary matrix, then $\det(E^\top) = \det(E)$.

Proof. Let E_1 be a type 1 elementary matrix formed by interchanging two rows of the identity matrix I . Let us take two elements in row r_1 and r_2 and columns c_1 and c_2 respectively which are ones, and if they are swapped then,

$$E_1 = \begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 & \dots & 0 \\ 0 & 0 & \dots & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 0 & \dots & 1 \end{bmatrix} \begin{array}{l} \\ \leftarrow \text{row } r_1 \\ \leftarrow \text{row } r_2 \\ \\ \\ \end{array}$$

Hence, $(E_1)_{ij} = 1$ when $i = j$ except for $i = r_1, j = r_1$ and $i = r_2, j = r_2$, and $(E_1)_{ij} = 1$ when $i = r_1, j = r_2$ and $i = r_2, j = r_1$, and $(E_1)_{ij} = 0$ at all other positions. Now if we take transpose of E_1 , then we get $(E_1^\top)_{ij} = 1$ when $i = j$ except for $i = r_1, j = r_1$ and $i = r_2, j = r_2$, and $(E_1^\top)_{ij} = 1$ when $i = r_2, j = r_1$ and $i = r_1, j = r_2$, and $(E_1^\top)_{ij} = 0$ at all other positions which is equal to E_1 . Thus, $E_1^\top = E_1$, and taking determinant both sides,

$$\det(E_1^\top) = \det(E_1).$$

Let E_2 be a type 2 elementary matrix formed by multiplying one of the rows of identity matrix I with some scalar k . Let some row r of I be multiplied by k , then,

$$E_2 = \begin{bmatrix} 1 & 0 & \dots & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & k & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & \dots & 1 \end{bmatrix} \leftarrow \text{row } r$$

Since all off-diagonal elements are 0, then $E_2^\top = E_2$. Hence

$$\det(E_2^\top) = \det(E_2).$$

Now let E_3 be a type 3 elementary matrix formed by adding a nonzero multiple, say k of row r_1 of an identity matrix to row r_2 of the same identity matrix. Thus by Theorem 2.4, $\det(E_3) = 1$. Now $E_3^\top$ is a matrix where $(E_3^\top)_{ij} = 1$ for $i = j$ and all off diagonal elements are zeroes except $(E_3^\top)_{r_1, r_2} = k$. Now solving for the determinant of $E_3^\top$ for the row r_1 using the Theorem 2.2, we get

$$\begin{aligned} \det(E_3^\top) &= \sum_{j=1}^n (-1)^{r_1+j} (E_3^\top)_{r_1, j} \det((\tilde{E}_3^\top)_{r_1, j}) \\ &= (-1)^{r_1+r_1} (E_3^\top)_{r_1, r_1} \det((\tilde{E}_3^\top)_{r_1, r_1}) + (-1)^{r_1+r_2} (E_3^\top)_{r_1, r_2} \det((\tilde{E}_3^\top)_{r_1, r_2}) \\ &= \det((\tilde{E}_3^\top)_{r_1, r_1}) + (-1)^{r_1+r_2} .k. \det((\tilde{E}_3^\top)_{r_1, r_2}) \end{aligned}$$

Now since $(\tilde{E}_3^\top)_{r_1, r_1}$ is same as an identity matrix, thus $\det((\tilde{E}_3^\top)_{r_1, r_1}) = 1$. Also since $(\tilde{E}_3^\top)_{r_1, r_2}$ is a matrix containing only zeroes at the row r_2 , thus by Corollary 2.1.1, $\det((\tilde{E}_3^\top)_{r_1, r_2}) = 0$. Hence,

$$\det(E_3^\top) = 1 = \det(E_3)$$

□

Following are the facts about determinants:

1. If E is an elementary matrix obtained by interchanging two rows of I , then $\det(E) = -1$.
2. If E is an elementary matrix obtained by multiplying some row of I by the nonzero scalar k , then $\det(E) = k$.
3. If E is an elementary matrix obtained by adding a multiple of some row of I to another row, then $\det(E) = 1$.

Theorem 2.7. For any $A, B \in M_{n \times n}(F)$, $\det(AB) = \det(A) \cdot \det(B)$.

Proof. We begin by establishing the result when A is an elementary matrix. If A is an elementary matrix obtained by interchanging two rows of I , then $\det(A) = -1$. But then AB becomes a matrix obtained by interchanging two rows of B . Hence by Theorem 2.3,

$$\det(AB) = -\det(B) = \det(A) \cdot \det(B).$$

Now, if A is an elementary matrix obtained by multiplying some row of A by some nonzero scalar k , then $\det(A) = k$. But then AB becomes a matrix obtained by multiplying some row of B by the nonzero scalar k . Hence, by Theorem 2.1,

$$\det(AB) = k \det(B) = \det(A) \cdot \det(B).$$

Now if A is an elementary matrix obtained by adding a multiple of some row of A to another row, then $\det(A) = 1$. Then AB becomes a matrix obtained by adding a multiple of some row of B to another row. Hence by Theorem 2.4,

$$\det(AB) = \det(B) = 1 \times \det(B) = \det(A) \times \det(B).$$

Again, if A is an $n \times n$ matrix with rank less than n , then by Corollary 2.4.1, $\det(A) = 0$. We know $\text{rank}(AB) \leq \text{rank}(A) < n$. Similarly $\det(AB) = 0$. Thus

$$\det(AB) = \det(A) \cdot \det(B).$$

On the other hand, if A has rank n , then A is invertible and hence A can be written as a product

of elementary matrices, say, $A = E_m E_{m-1} \dots E_1$. Then by the first part of this proof,

$$\begin{aligned}
\det(AB) &= \det(E_m E_{m-1} \dots E_2 E_1 B) \\
&= \det(E_m) \cdot \det(E_{m-1} \dots E_2 E_1 B) \\
&= \dots \\
&= \det(E_m) \cdot \det(E_{m-1}) \dots \det(E_2) \cdot \det(E_1) \cdot \det(B) \\
&= \det(E_m E_{m-1} \dots E_2 E_1) \cdot \det(B) \\
&= \det(A) \cdot \det(B).
\end{aligned}$$

□

Corollary 2.7.1. *A matrix $A \in M_{n \times n}(F)$ is invertible if and only if $\det(A) \neq 0$. Furthermore, if A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.*

Proof. If $A \in M_{n \times n}(F)$ is not invertible then the rank of A is less than n . So $\det(A) = 0$ by Corollary 2.4.1. On the other hand, if A is invertible then by Theorem 2.7,

$$\det(A) \cdot \det(A^{-1}) = \det(AA^{-1}) = \det(I) = 1.$$

Hence $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$.

□

Till now we have only been studying the row operations on the matrices and their effect on the determinants, but what happens when we try to apply column operations

Theorem 2.8. *For any $A \in M_{n \times n}(F)$, $\det(A^t) = \det(A)$.*

Proof. If A is non-invertible then $\text{rank}(A) < n$. We know, $\text{rank}(A^t) = \text{rank}(A) < n$, and so A^t is not invertible and by Corollary 2.4.1,

$$\det(A^t) = 0 = \det(A).$$

On the other hand, if A is invertible then A can be written as a product of elementary matrices, say $A = E_m \dots E_2 E_1$. Since $\det(E^t) = \det(E)$ from Theorem 2.6, then by Theorem 2.7,

$$\begin{aligned}
\det(A^t) &= \det(E_1^t E_2^t \dots E_m^t) \\
&= \det(E_1^t) \cdot \det(E_2^t) \dots \det(E_m^t) \\
&= \det(E_1) \cdot \det(E_2) \dots \det(E_m) \\
&= \det(E_m) \dots \det(E_2) \cdot \det(E_1) \\
&= \det(E_m \dots E_2 E_1) \\
&= \det(A).
\end{aligned}$$

□

This proof shows us that it doesn't matter if we apply row operations or column operations on the matrices, they give us the same results related to determinants.

3 Conclusion

In this document, we have systematically established the foundational properties of the determinant scalar-valued function on $M_{n \times n}(F)$. By employing mathematical induction, we showed that the determinant behaves as an n -linear alternating form with respect to the rows of a matrix. Key results include the equivalence of cofactor expansion along any arbitrary row, the multiplicative property $\det(AB) = \det(A)\det(B)$, and the invariance of the determinant under transposition ($\det(A^T) = \det(A)$), which validates the applicability of all established properties to column operations as well. Crucially, the non-vanishing of the determinant ($\det(A) \neq 0$) serves as a complete algebraic characterization of matrix invertibility and full row rank. These structural properties lay the essential groundwork for further study in linear algebra, including diagonalizability and canonical forms.

References

- [1] Friedberg, S. H., Insel, A. J., & Spence, L. E. (2018). *Linear Algebra* (5th ed.). Pearson.